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PRACTICE AND TEACHING

# Practicing the Encounter

*Artificial intelligence in the training of psychotherapists and counselors: simulated clients, skills feedback, and supervision*

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ABSTRACT

Training in psychotherapy and counseling has long relied on role-plays, recorded sessions, and supervision, yet structured opportunities for repeated practice with specific feedback remain scarce. Advances in natural language processing and large language models have produced two families of tools that address this gap: simulated clients with whom trainees can rehearse conversations, and automated systems that code utterances and return feedback on skills such as open questions, reflections, and empathic responding. This practice-oriented article outlines the didactic rationale for such tools within the framework of deliberate practice, summarizes the available evidence from feasibility studies and randomized trials with nontherapist and peer samples, and illustrates their use in a constructed teaching vignette. It then discusses their limits: simulated clients may be implausibly cooperative, stereotyped, or inconsistent; automated feedback captures countable behaviors more readily than relational quality; and recordings raise questions of consent and data protection. From a meaning-centered perspective, the article argues that AI tools can usefully train technique but cannot replace supervised encounters with real persons. Recommendations for curriculum integration, transparency, and evaluation are offered.

**Keywords:** artificial intelligence · psychotherapy training · simulated clients · deliberate practice · clinical supervision

## 1 Introduction

Learning to conduct psychotherapeutic conversations is an apprenticeship in a demanding interpersonal skill. Trainees must learn to listen, to ask open questions, to reflect meaning and emotion, to tolerate silence, and to respond to distress without losing their footing. Traditional curricula teach these competencies through lectures, peer role-plays, observation of recorded sessions, and supervised casework. Each format has its value, but all share a practical limitation: opportunities to practice a specific skill repeatedly, with timely and specific feedback, are scarce and costly (Tanana et al., 2019).

Over the last decade, computational methods have begun to address this bottleneck. Research groups have developed systems that automatically code therapist utterances in motivational interviewing (Atkins et al., 2014; Flemotomos et al., 2022), feedback tools that summarize such codes for clinicians (Imel et al., 2019), and conversational agents that play the role of a client (Tanana et al., 2019). Large language models have broadened these possibilities considerably (Stade et al., 2024). This article examines what these tools can contribute to training, where their limits lie, and how they might be integrated into curricula in a way that remains faithful to the relational and existential core of psychotherapy.

## 2 Practice, Feedback, and the Development of Expertise

The didactic case for AI-supported training rests largely on the concept of deliberate practice: effortful, repeated rehearsal of well-defined subskills slightly beyond one's current level, accompanied by immediate feedback and opportunities for correction (Ericsson et al., 1993). Rousmaniere (2017) has adapted this concept to psychotherapy, proposing that therapists rehearse difficult moments, such as responding to hostility or to an expression of hopelessness, in brief and repeatable exercises rather than waiting for such moments to occur in actual sessions.

Two lines of evidence make this approach attractive. First, studies of clinical experience suggest

that therapists do not automatically become more effective over time; in a large naturalistic dataset, Goldberg et al. (2016) reported that client outcomes, on average, did not improve with therapist experience and even declined slightly. Second, research on training in motivational interviewing indicated that a workshop alone produced gains in proficiency that were modest and partly unstable, whereas adding feedback and coaching was associated with better skill retention (Miller et al., 2004). Feedback, in other words, is not an optional supplement but a central mechanism of learning, and it is precisely feedback that is in short supply.

### 3 Simulated Clients: From Scripted Role-Play to Language Models

Simulated clients have a long history in clinical education, including trained actors and standardized patients. Early computational versions were rule-based or retrieval-based. A well-documented example is ClientBot, a text-based agent trained on a large archive of psychotherapy transcripts that responds like a client and gives trainees feedback on their use of open questions and reflections (Tanana et al., 2019). In a randomized evaluation with nontherapist participants, those who practiced with immediate feedback used reflections considerably more often than a control group that received only education and encouragement, both during practice and after the feedback had been removed (Tanana et al., 2019).

Large language models make it possible to generate clients with detailed backstories, varying degrees of ambivalence, and different communication styles, and to adjust difficulty on demand. A trainee can, for example, ask the simulation to portray a client who responds to premature advice with withdrawal. Such flexibility is pedagogically valuable because it allows repeated exposure to clinically challenging situations that are rare, ethically delicate, or unpredictable in real practice.

The same flexibility creates new problems. Language models generate statistically plausible text without understanding or lived experience (Bender et al., 2021). Simulated clients may therefore be unrealistically articulate, cooperative, or quick to gain insight; they may reproduce cultural or diagnostic stereotypes present in their training data; and they may drift out of character over a longer conversation. Instructors should therefore treat simulated clients as exercises of known artificiality rather than as faithful models of patients.

### 4 Automated Feedback on Conversation Skills

A second family of tools analyzes what the trainee says. Building on behavioral coding systems for motivational interviewing, machine-learning models have been trained to classify utterances, for example as open or closed questions, simple or complex reflections, or responses consistent or inconsistent with motivational interviewing (Atkins et al., 2014). Later work combined speech recognition, speaker identification, and language models to estimate such codes and session-level indicators directly from audio recordings (Flemotomos et al., 2022). In a feasibility study, Imel et al. (2019) reported that an automated feedback report based on such codes could be delivered to counselors and was generally received as acceptable and understandable.

Related work has explored feedback on empathy. In a randomized study on a text-based peer-support platform, Sharma et al. (2023) reported that peer supporters who received AI-generated suggestions for rephrasing their responses produced conversations rated as more empathic than those of supporters without such assistance. Although peer supporters are not psychotherapists, the study illustrates a training-relevant mechanism: feedback delivered at the moment of formulation, which the supporter remains free to accept, adapt, or reject.

Automated feedback is most reliable for behaviors that can be counted and defined at the level of single utterances. Relational qualities such as timing, warmth, presence, or the capacity to stay with a client's despair are far harder to operationalize. There is a risk that what is measurable becomes what is taught: trainees may learn to increase the number of their reflections without deepening their understanding of the person in front of them.

### 5 A Constructed Teaching Vignette

The following vignette is fictional and was constructed for teaching purposes; it does not describe real persons or data. A second-year trainee in a counseling program practices with a simulated client based on a language model, described as a man in his fifties who has recently lost his job and says that nothing he does matters anymore. In the first run, the trainee responds quickly with reassurance and suggestions for job applications. The simulated client becomes terse. An automated feedback panel notes many closed questions and few reflections.

In a second run, the trainee slows down, reflects the client's sense of futility, and asks what used to give his

work significance. The simulation responds with more elaborate statements. In the subsequent group supervision, however, the supervisor does not simply praise the higher number of reflections. Instead, the group discusses what the exercise could not show: how the trainee's own discomfort with hopelessness drove the initial rush to advice, how a real client might have responded with silence or anger, and how a question about meaning can open a Socratic dialogue only when it is asked with genuine interest rather than as a technique (Frankl, 1969). The simulation served as material for reflection; the learning took place in the human conversation about it.

## 6 Didactic Limits, Supervision, and Ethical Safeguards

Several limits follow from the preceding discussion. The evidence remains preliminary: most studies involve nontherapist participants, peer supporters, or short periods of exposure, and outcomes are usually measured as skill behavior in practice tasks rather than as client outcomes in later treatment (Stade et al., 2024; Tanana et al., 2019). Transfer from simulation to real sessions has not been established.

Supervision cannot be delegated to these tools. Supervision is concerned not only with techniques but with the trainee's development as a person, with countertransference, ethical judgment, and responsibility for real clients. Automated feedback may enrich supervision by providing a shared and concrete basis for discussion, but it cannot replace the supervisor's judgment or the relationship within which professional identity is formed (Bennett-Levy, 2006).

Ethical and legal safeguards are also required. Recordings of real sessions used for automated feedback contain highly sensitive health information; clients must give informed consent, and data processing must comply with applicable data-protection law. Commercial tools should be examined with regard to where data are stored and whether they are used to train models. Trainees should be informed about the artificiality and possible biases of simulated clients, and programs should evaluate their tools rather than assume their effectiveness.

From a meaning-centered perspective, a further point deserves emphasis. Logotherapy understands the therapeutic encounter as a meeting between persons in which the therapist accompanies the client's search for meaning without prescribing it (Frankl, 1969). A simulated client has no meaning to find and no freedom to take a stand. Practicing with simulations can sharpen attention and technique, but the existential dimension of the encounter can be learned only in contact with real persons and through reflection on one's own attitude toward suffering.

## 7 Conclusion

AI-based simulated clients and automated feedback systems offer a promising way to expand deliberate practice in the training of psychotherapists and counselors. Their limits lie in their artificiality, in a bias toward the measurable, in an evidence base that is still thin, and in privacy risks. Programs that integrate them should do so transparently, as a complement to role-play, casework, and supervision, and should evaluate their effects on competence and, ultimately, on client care. Used in this way, the machine can become a practice partner, while the encounter remains a human task.

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